

**Enhancing the Co-op Map through Visual Data Exploration
Client Project – Mercury Co-operative Limited**

Macquarie University | 2026
Group Project

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1. INTRODUCTION

This project aims to improve and extend the existing Co-op Map designed by Mercury. While the current map is useful for locating cooperatives and viewing basic information such as names and addresses, the proposed application makes the map more interactive and exploratory for user decision-making needs by supplementing more data visuals and features. The proposed environment enables users to filter, compare and interpret data based on their own purposes and reasoning instead of presenting fixed conclusions.

The envisaged value for Mercury is that the application allows users to explore more decision-supportive insights beyond the current map through interactive filtering and comparison. In particular, it supports exploration of cooperative development opportunities, possible collaboration patterns among existing cooperatives, and areas where cooperative models may support workforce training or employment pathways.

The project also extends the map's usage to additional stakeholder groups, including education employability teams and people experiencing housing stress. For people working in employability teams at educational institutions, the application can support the identification of meaningful student employability partnerships with cooperatives through innovation alignment, shared values and accessibility. For people experiencing housing stress, the application can support the exploration of housing cooperatives as an affordable housing option through the reported benefits, existing housing co-op options and potential areas where future housing cooperatives may be established.

Through three personas, the application shows how the Co-op Map can support different purposes of visual data exploration. Overall, the proposed application is intended to make the Co-op Map more interactive and user-centred by allowing users to explore data based on their own purposes, filters and reasoning. The insights generated may vary across users because they can apply different filters, focus on different variables, and interpret the visual patterns alongside other factors within their own problem space. Therefore, the application does not aim to direct users toward a particular decision, but to empower a more informed, data-driven decision-making process.

2. DESIGN TEMPLATES FOR THREE PERSONAS

Persona 1 - Board Director Peter

Persona	‘Board Director Peter’
Role	A board director of Mercury Co-operative
Context	A senior leader of Mercury who is engaged in value-creating opportunities for Mercury’s members and Mercury’s ‘parent organisation’ The Co-op Federation
Key activities/ decision making needs and interests	Interested in extending the Co-op Map and turning it into a visual data exploration environment of the Australian Cooperatives. The existing Co-op Map is seen as a good start. The visual data exploration environment will enable Mercury’s members as well as other interested stakeholders (from The Co-op Federation and beyond) to learn about Australian member organisations and their impact, connect similar cooperatives, collaborate, explore future opportunities and so on.
DV Questions	Question 1: How does industry instability reveal opportunities for cooperative development? Question 2: How can similarities between cooperatives reveal potential collaboration or partnership opportunities? Question 3: How do regions with both unemployment pressure and labour shortages reveal opportunities for cooperative training, employment, or workforce-matching models?

Persona 2 - MQ Employability Manager Min

Persona	‘MQ Employability Manager Min’
Role	A leader of MQ Employability and Graduate Success
Context	Min supports MQ students and graduates' career paths by connecting them with career development, industry engagement, and employment opportunities.
Key activities/ decision making needs and interests	Interested in exploring new opportunities for MQ students and graduates in the Australian cooperative sector. The visual data exploration environment will enable Min to identify which cooperatives are most suitable for MQ employability partnerships and determine whether partnership opportunities are practically viable for students seeking internships, graduate roles, industry projects, or volunteering experiences.
DV Questions	Question 1: How can Min explore innovation-oriented student partnerships using precinct proximity and industry alignment? Question 2: How do cooperatives’ ICA classifications indicate potential alignment with university values and student-development goals? Question 3: How can Min explore cooperatives in selected sectors that are accessible from Macquarie University?

Persona 3 - Young Adult Renter Lucky

Persona	‘Young Adult Renter Lucky’
Role	A young adult renter experiencing housing stress and exploring affordable housing alternatives
Context	Lucky is a medium-income renter whose private rental costs take up a large share of his income. He is not in an extreme housing crisis, but he wants a more affordable, stable, and community-oriented housing option. Lucky is interested in understanding whether housing cooperatives could offer a realistic pathway to reduce housing stress, either by joining an existing housing cooperative or by exploring suburbs where a new cooperative could be developed.
Key activities/ decision making needs and interests	Interested in exploring the Australian housing cooperative landscape to determine whether cooperative housing is a realistic solution to his housing stress. The visual data exploration environment will enable Lucky to evaluate cooperative housing as a viable option, identify existing housing cooperatives that match his practical preferences, and explore areas where a new housing cooperative could be established.
DV Questions	Question 1: What benefits could Lucky consider when exploring housing cooperatives as a response to housing stress? Question 2: How well do existing housing cooperatives match Lucky’s preferences and practical needs? Question 3: How suitable are different suburbs for establishing a new housing cooperative, based on housing affordability pressure, limited housing options, and liveability?

3. METADATA OVERVIEW OF PROJECT DATASETS

Data sample 1: Metadata of *80_selected_cooperatives*

Sheet Name	Column / Field	Definition
80_selected_cooperatives	Sheet Overview	Subset of 80 Australian cooperatives selected from Mercury's full Co-op Directory. Primary dataset for Persona 1 (Peter) and Persona 2 (Min).
80_selected_cooperatives	ID	Unique numeric identifier assigned by Mercury for each co-operative in the directory.
80_selected_cooperatives	cooperative_name	Registered legal name of the cooperative.
80_selected_cooperatives	state	Australian state or territory where the co-operative is registered.
80_selected_cooperatives	state_code	State abbreviation matching the state column.
80_selected_cooperatives	city	Suburb or town where the co-operative is located.
80_selected_cooperatives	street	Street address of the co-operative.
80_selected_cooperatives	latitude	Geographic latitude coordinate (decimal degrees).
80_selected_cooperatives	longitude	Geographic longitude coordinate (decimal degrees).
80_selected_cooperatives	ica1	Binary indicator: alignment with ICA Principle 1 - Voluntary and Open Membership.
80_selected_cooperatives	ica2	Binary indicator: alignment with ICA Principle 2 - Democratic Member Control.
80_selected_cooperatives	ica3	Binary indicator: alignment with ICA Principle 3 - Member Economic Participation.
80_selected_cooperatives	ica4	Binary indicator: alignment with ICA Principle 4 - Autonomy and Independence.
80_selected_cooperatives	ica5	Binary indicator: alignment with ICA Principle 5 - Education, Training, and Information.
80_selected_cooperatives	ica6	Binary indicator: alignment with ICA Principle 6 - Cooperation Among Cooperatives.
80_selected_cooperatives	ica7	Binary indicator: alignment with ICA Principle 7 - Concern for Community.
80_selected_cooperatives	sector_division	Single-letter ANZSIC 2006 industry division code (A-S).
80_selected_cooperatives	sector	Full name of ANZSIC industry division.
80_selected_cooperatives	profit_type	Profit orientation of the cooperative.
80_selected_cooperatives	commodity_section	Numeric code for commodity classification.
80_selected_cooperatives	commodity_type	Commodity category name (e.g., "Community, Social and Personal Services").

Data sample 2: Metadata of *survival_by_industry*

Sheet Name	Column / Field	Definition
survival_by_industry	Sheet Overview	Business entry, exit and survival rates by ANZSIC industry sector. Used in Persona 1 Question 1 to identify industries with instability that may benefit from co-operative structures.
survival_by_industry	sector	ANZSIC industry division name.
survival_by_industry	entry_rate_%	Rate of new business entries during the reference period: (Entries Total / Businesses Operating at start) × 100
survival_by_industry	exit_rate_%	Rate of business exits during the reference period: (Exits Total / Businesses Operating at start) × 100
survival_by_industry	survival_rate_of_entries_%	Percentage of businesses that entered the market in 2021 and were still operating by June 2025: (Survived to June 2025 / Survived to June 2021) × 100

Data sample 3: Metadata of *nsw_cooperatives_precincts*

Sheet Name	Column / Field	Definition
nsw_cooperatives_precincts	Sheet Overview	NSW innovation precincts (12) and 25 NSW co-operatives from the 80-sample, with calculated proximity between each co-op and its nearest precinct, plus distance from Macquarie University. Used in Persona 2 Questions 1 and 3.
nsw_cooperatives_precincts	entity_name	Name of the precinct or co-operative.
nsw_cooperatives_precincts	entity_type	Indicates whether the row is a precinct or a co-operative.
nsw_cooperatives_precincts	precinct_filter	Name of the nearest aligned innovation precinct for each co-op (for precincts, this matches their own name).
nsw_cooperatives_precincts	latitude	Geographic latitude (decimal degrees).
nsw_cooperatives_precincts	longitude	Geographic longitude (decimal degrees).
nsw_cooperatives_precincts	proximity_km	Straight-line distance from the co-operative to its nearest aligned innovation precinct in kilometres (N/A (not applicable) for precinct rows and for co-operatives with no aligned sampled precinct).
nsw_cooperatives_precincts	ica1 to ica7	Same ICA binary indicators as in 80_selected_cooperatives sheet, copied for co-op rows only (N/A (Not applicable) for precinct rows because ICA coding was only applied to co-operatives).
nsw_cooperatives_precincts	sector	ANZSIC sector of the co-operative (N/A (not applicable) for precinct rows because the sector classification was only applied to co-operatives here).
nsw_cooperatives_precincts	distance_from_mq	Distance from the entity (co-op or precinct) to Macquarie University main campus (Macquarie Park) in kilometres.

The screenshots above provide examples of the metadata used to document selected project datasets. The full underlying datasets are not included in this public portfolio version.

4. VISUAL ANSWERS FOR PERSONAS' QUESTIONS

4.1. Persona 1: Board Director Peter

Question 1: How does industry instability reveal opportunities for cooperative development?

Visual answer:

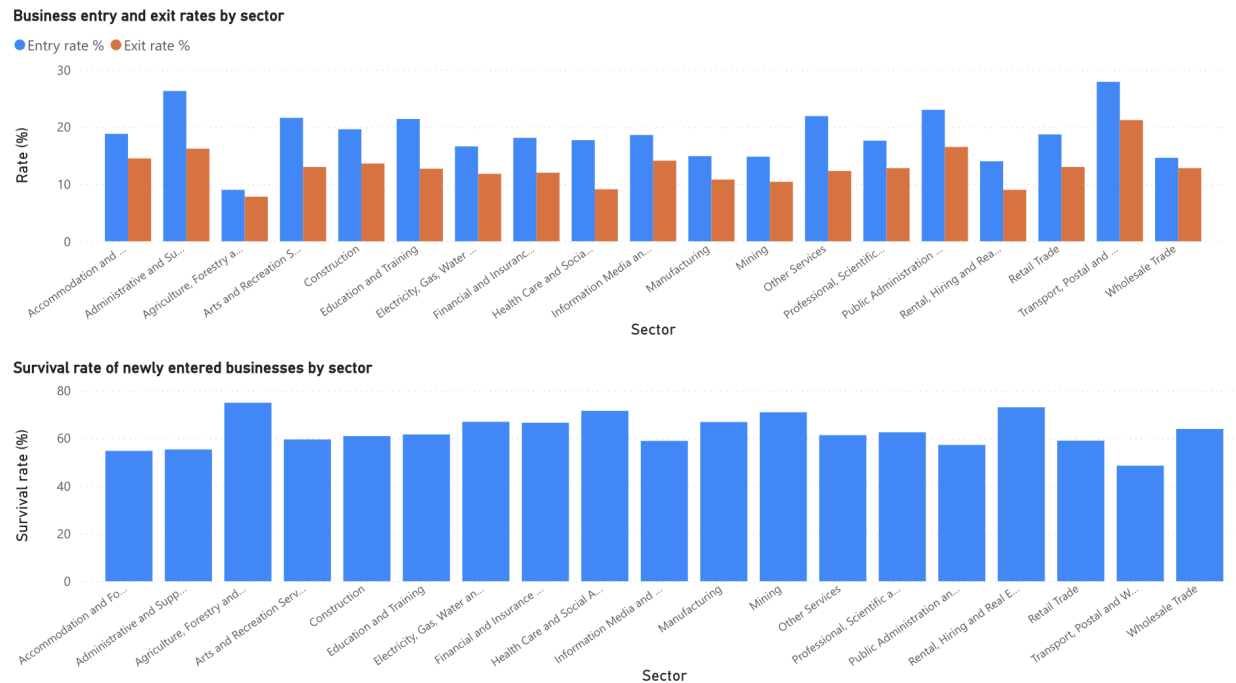


Figure 1. Visual answer showing business entry, exit and new business survival patterns by sector.

Figure 1 helps Peter explore industry instability by comparing business entry rates, exit rates and survival rates across sectors. Higher entry and exit rates may indicate sectors with greater uncertainty, while lower survival rates highlight sectors with weaker business continuation patterns. Together, these indicators support Peter in identifying sectors where cooperative models may offer a pathway to greater stability and resilience for businesses.

Question 2: How can similarities between cooperatives reveal potential collaboration or partnership opportunities?

Visual answer:

State ☐ Select all ☐ Australian Capital Territory ☐ New South Wales ☐ Queensland ☐ South Australia ☐ Tasmania ☐ Victoria ☐ Western Australia	Sector ☐ Select all ☐ Accommodation and Food Services ☐ Agriculture, Forestry and Fishing ☐ Arts and Recreation Services ☐ Construction ☐ Education and Training ☐ Electricity, Gas, Water and Waste Services ☐ Health Care and Social Assistance	Commodity type ☐ Select all ☐ Agriculture, Forestry and Fishery Products ☐ Business Services; Agricultural, Mining and Manufacturing Services ☐ Community, Social and Personal Services ☐ Construction ☐ Construction Work and Constructions; Land ☐ Food Products, Beverages and Tobacco ☐ Food Products, Beverages and Tobacco; Textiles, Apparel and Leather Products	Profit type ☐ Select all ☐ For-profit ☐ Not-for-profit	ICA ☐ Select all ☐ ica1 ☐ ica2 ☐ ica3 ☐ ica4 ☐ ica5 ☐ ica6 ☐ ica7
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Co-operative collaboration map

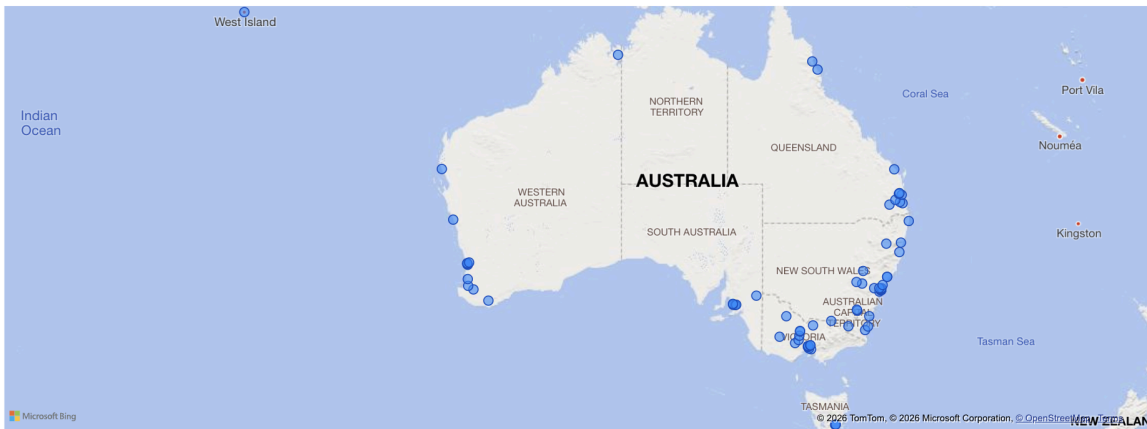


Figure 2. Visual answer showing cooperatives by location and profile for collaboration exploration.

Figure 2 allows Peter to explore cooperatives that share similar characteristics, such as industry, location, function, profit status or ICA principle alignment. This supports the identification of possible collaboration or partnership opportunities, as cooperatives with similar profiles may share common operational challenges, member needs or strategic interests.

Question 3: How do regions with both unemployment pressure and labour shortages reveal opportunities for cooperative training, employment, or workforce-matching models?

Visual answer:

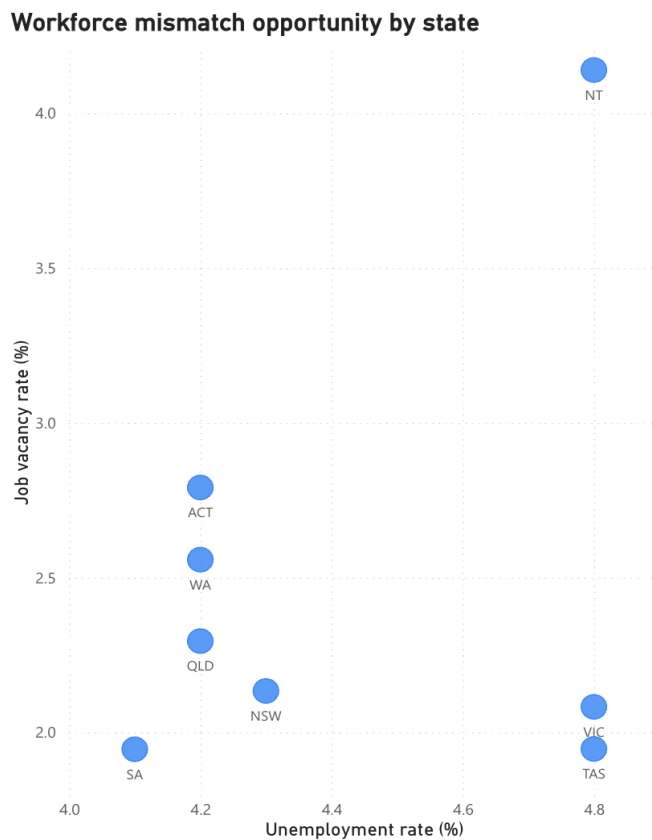


Figure 3. Visual answer showing unemployment pressure and job vacancy rate by state.

Figure 3 compares unemployment rate and job vacancy rate by state to identify potential workforce mismatch. States with both relatively high unemployment and high job vacancy rates may suggest that available workers are not being effectively matched with labour demand. This can help Peter explore where cooperative workforce solutions may be relevant, such as worker cooperatives, training pathways, employment programs or shared staffing models.

4.2. Persona 2: MQ Employability Manager Min

Question 1: How can Min explore innovation-oriented student partnerships using precinct proximity and industry alignment?

Visual answer:

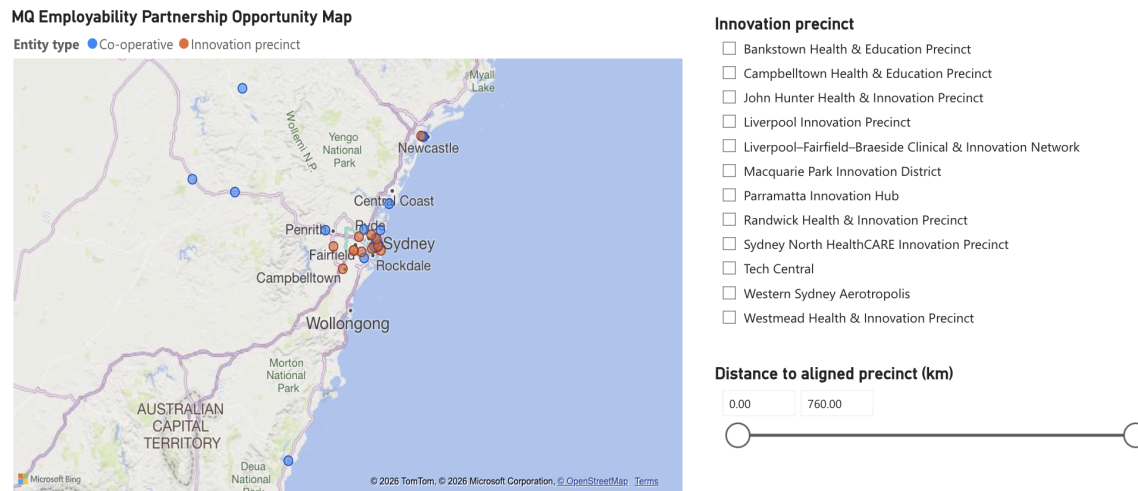


Figure 4. Visual answer showing cooperatives and innovation precincts for employability partnership exploration.

Figure 4 helps Min explore cooperatives in relation to innovation precincts. The map allows users to view cooperatives and precincts together, while the precinct and distance filters allow Min to narrow the search to cooperatives located closer to selected aligned precincts. This supports the identification of cooperatives that may be suitable for innovation-oriented student partnership opportunities, particularly where the cooperative's sector aligns with the focus of a nearby precinct.

Question 2: How do cooperatives' ICA classifications indicate potential alignment with university values and student-development goals?

Visual answer:

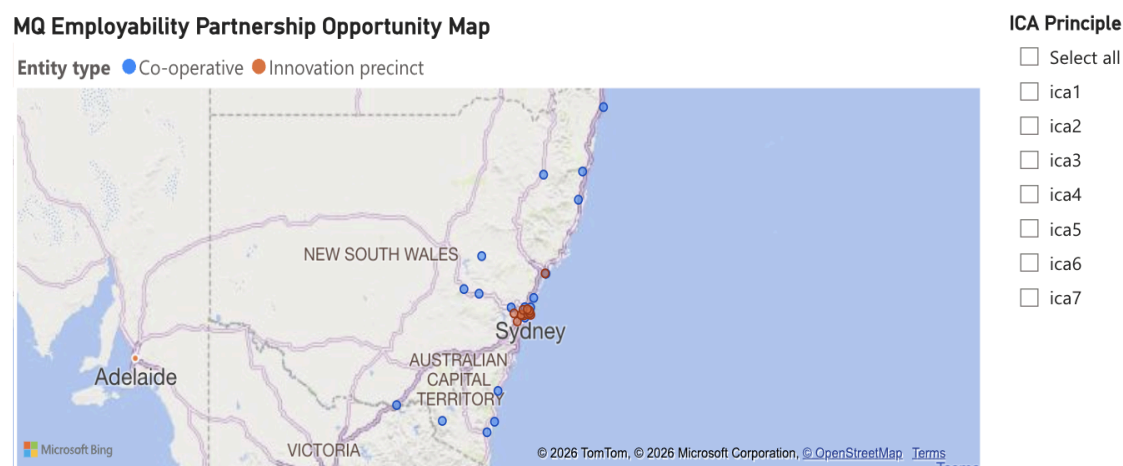


Figure 5. Visual answer showing ICA principle filtering for values-aligned cooperative partnerships.

Figure 5 supports Min in exploring potential value alignment through ICA principles. This helps Min identify cooperatives that may share values relevant to student development, ethical practice and community engagement, and may therefore be suitable for employability partnerships.

Question 3: How can Min explore cooperatives in selected sectors that are accessible from Macquarie University?

Visual answer:

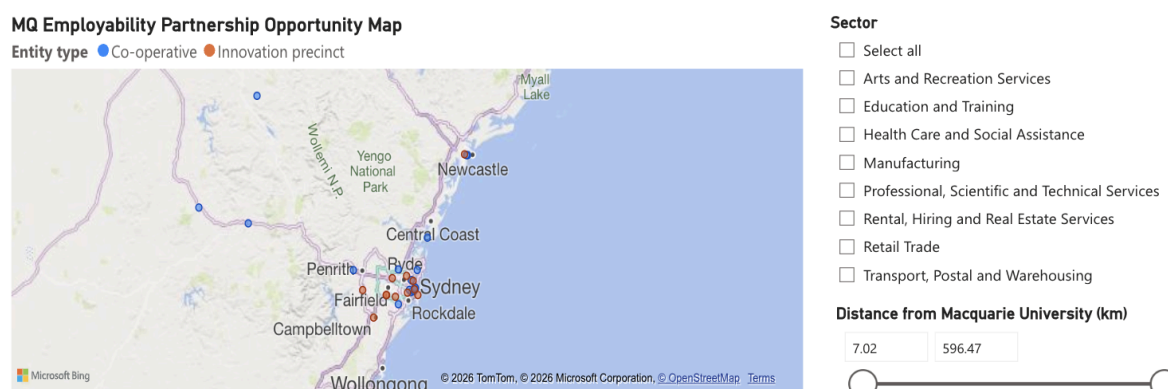


Figure 6. Visual answer showing cooperatives by distance and sector from Macquarie University.

Figure 6 allows Min to explore cooperatives based on both sector relevance and geographic accessibility from Macquarie University. The sector filter helps identify cooperatives that may match student interests or study areas, while the distance filter helps determine whether partnership opportunities are practical for students to access.

4.3. Persona 3: Young Adult Renter Lucky

Question 1: What benefits could Lucky consider when exploring housing cooperatives as a response to housing stress?

Visual answer:

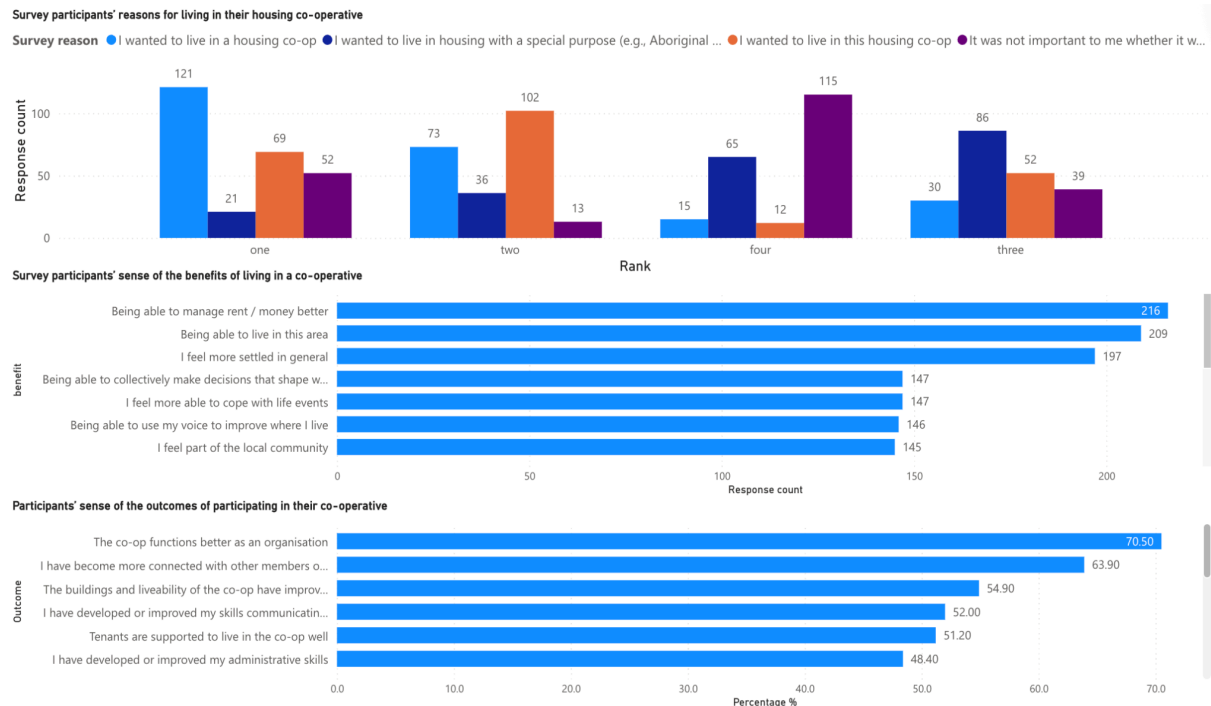


Figure 7. Visual answer showing reported reasons, benefits and outcomes of housing cooperative living.

Figure 7 helps Lucky explore why people choose housing cooperatives and what benefits they may experience. The reasons chart shows common motivations for living in housing cooperatives, while the benefits and outcomes visuals show reported improvements such as better stability, stronger community connection, participation in decision-making and improved living experience. This allows Lucky to consider whether these reported benefits are relevant to his own housing needs.

Question 2: How well do existing housing cooperatives match Lucky’s preferences and practical needs?

Visual answer:

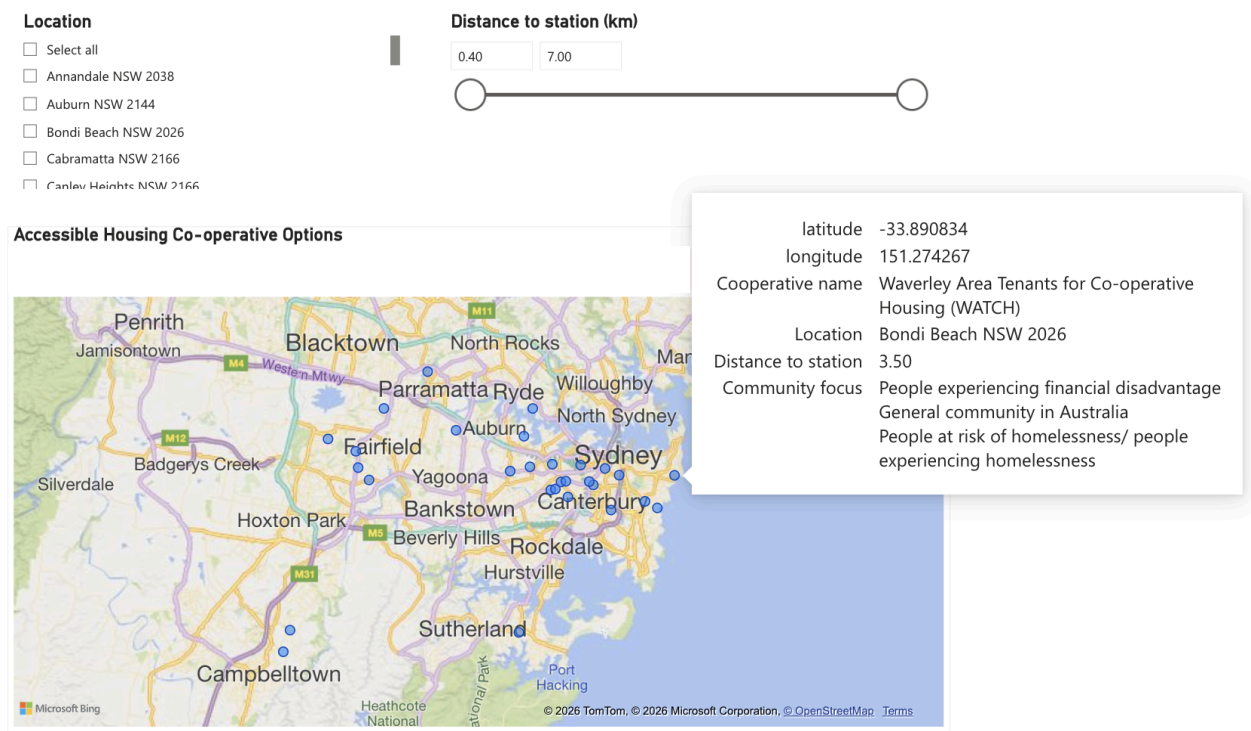


Figure 8. Visual answer showing accessible housing cooperative options by location, station distance and tooltip details.

Figure 8 supports Lucky in exploring existing housing cooperative options based on practical preferences. The map shows housing cooperative locations, while the location and distance-to-station filters allow Lucky to narrow the options by geographic area and public transport accessibility. Tooltips also provide additional information about each cooperative. This helps Lucky compare existing housing cooperative options in a way that reflects practical housing decision-making.

Question 3: How suitable are different suburbs for establishing a new housing cooperative, based on housing affordability pressure, limited housing options, and liveability?

Visual answer:

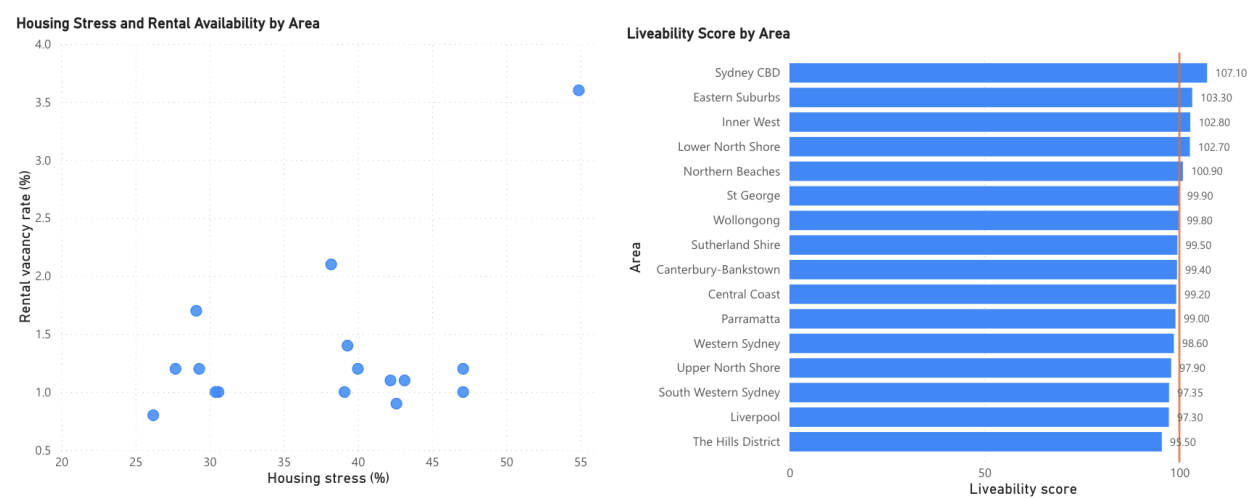


Figure 9. Visual answer showing area suitability indicators for future housing cooperative establishment.

Figure 9 helps Lucky explore potential areas for future housing cooperative development by comparing housing stress, rental vacancy rate and liveability. Areas with higher housing stress may indicate affordability pressure, while lower rental availability may suggest limited housing options. The liveability score adds another layer by showing how well an area is connected to practical daily needs, such as work, education, transport and amenities.

5. ANALYSIS OF DATA QUALITY, DATA AND VISUAL ETHICS AND UNINTENDED DATA HARM

5.1. Data Quality

This project considered data quality through three main dimensions: syntactic, semantic and pragmatic data quality. Syntactic data quality refers to whether the data is technically usable by the computer. In this project, this was important because the datasets needed to work in Power BI for maps, filters, tooltips and visual comparisons. Semantic data quality refers to whether the meaning of the data is clear. This was addressed by establishing a clear metadata set, so that users can understand what each indicator actually represents. Pragmatic data quality refers to whether the data is suitable for the intended use. The indicators were selected based on whether they helped the three personas answer their questions.

Several data quality processes were used throughout the project. First, data harvesting was conducted by collecting information from the main Co-op Map dataset, public datasets, websites and manually gathered records. Filtering was then applied to remove unnecessary fields and keep only variables relevant to the personas' questions. Data cleansing was used to correct inconsistent formats, rename unclear columns and remove invalid records. Validation was also performed by checking data records against original sources where possible and questioning indicators that were unclear or difficult to interpret. These processes mainly improved syntactic and semantic data quality.

The project also involved integration and consolidation. Different datasets were combined into final tables that could be imported into Power BI. This required aligning formats, units, locations and categories so that the datasets could work together in the dashboard. Analysis was then used to transform raw or unstructured information into usable indicators, such as proximity marks, ICA principle coding, housing stress measures and opportunity signals. Presentation and visualisation supported pragmatic data quality by making the data easier for users to explore through charts, maps, filters and tooltips. Customisation was also added through filters by sector, region, precinct, distance or attributes relevant to personas, allowing users to generate different insights depending on their own purposes.

However, several limitations remain. Some data was manually gathered, such as approximate distances from Google Maps, ICA principle classifications and industry coding. Although these were checked carefully, they may still involve a degree of judgement. Some proxy indicators also do not perfectly measure the concept they represent. For example, proximity to an innovation precinct suggests possible alignment, but does not prove partnership readiness. Similarly, housing indicators can suggest areas with potential, but cannot prove that a housing cooperative can be established there. Most of the datasets were also based on selected samples rather than broad coverage. Therefore, the proposed application should only be understood as an example of a visual exploration environment that supports decision-making needs, not as a tool that produces final or definitive decisions.

5.2. Data Ethics

This project considered data ethics through the PAPA framework: privacy, accuracy, property and accessibility. The main ethical concerns were related to approximate housing cooperative locations, manually coded indicators, source acknowledgement and the level of data detail made accessible to users.

Privacy:

The main privacy concern relates to housing cooperative locations. The dashboard does not display exact street addresses. Instead, it uses approximate pins based on the Common Equity housing locations map to support general geographic exploration. However, because the pins may still be close to actual housing cooperative sites, they may still carry some privacy implications.

Accuracy:

The dashboard uses several manually gathered or interpreted indicators, which may affect data accuracy. For example, ICA classifications and industry sectors were coded from cooperative website information, while some distance measures were manually estimated using Google Maps. These indicators were checked as carefully as possible, but they may still involve a degree of judgement. Therefore, measures such as proximity to an innovation precinct should be interpreted as possible alignment, not proof of partnership readiness or confirmed opportunity.

Property:

The project uses data from the main Co-op Map dataset, public datasets, public websites and published survey figures. These sources remain owned by their original providers and should be properly acknowledged in the reference section below. The project does not claim ownership over the original datasets.

Accessibility:

The dashboard is designed for exploration, but not all levels of data detail should be equally accessible. For example, approximate longitudes and latitudes of housing cooperatives may carry greater privacy risk if made visible to all users. Therefore, the dashboard should provide enough information for meaningful exploration while avoiding unnecessary exposure of underlying datasets. Definitions, assumptions and limitations should also remain accessible so users can interpret the visuals responsibly.

5.3. Visual Ethics

Visual ethics was considered to reduce the risk that users may interpret the dashboard visuals as stronger or more certain than the data can support. Since the project is designed as a visual data exploration environment, the visuals should help users explore patterns without creating misleading impressions or directing them toward a fixed conclusion.

To reduce visual ethics risks, the dashboard used relatively simple charts, maps, cards and filters that are easier for users to understand. Labels, figure titles and tooltips were added to clarify what each visual shows. We also avoided unnecessary visual features that could exaggerate the data, make differences appear more extreme, or distract users from the intended meaning of the visual.

Several potential visual ethics issues still need to be considered. For the MQ Employability Partnership Opportunity Map, filtering cooperatives by proximity or alignment to a selected precinct may create a partial story. Cooperatives that do not appear after filtering might be interpreted as having no possible opportunity. Instead, they may only be located beyond the selected distance threshold or not captured by the simplified proxy used in this dashboard. Also, in the innovation precinct visual, one pin is used to represent each innovation precinct, although a precinct may cover a larger area. Therefore, the measured distance between a cooperative and a precinct should be carefully interpreted as an approximate proximity indicator, not an exact measure of access to the whole precinct. For Lucky's housing cooperative map, the approximate pins could be misread as exact residential locations, even though they are intended only as area indicators.

If Mercury continues to develop the application, it should regularly review how filters, colours, labels and visual positioning affect user interpretation. Data legends, definitions and limitations should remain visible, and filtered results should not be presented as final conclusions. The dashboard should also be tested with users to identify whether any visual creates confusion or misleading assumptions.

5.4. Application of Do No Harm Principles and Unintended Harm Considerations

The project applied relevant Do No Harm principles to reduce the chance that the dashboards unintentionally stereotype, exclude or mislead users. People-first and respectful language was used throughout the datasets and visuals. For example, the project used wording such as “people experiencing housing stress” and “people experiencing financial disadvantage” rather than labels that may define people mainly by disadvantage or demographic characteristics.

The project also considered colour, ordering and accessible language. Light-to-dark colour scales were avoided for demographic or community groups, and colour choices were kept as neutral as possible to avoid implying that some groups, suburbs or cooperatives are “better” or “worse”. Data entries in charts were ordered purposefully, such as by count or alphabetically, rather than simply following the raw dataset order. The dashboard also used plain language variable names, tooltips and metadata so that non-technical users could understand what each indicator means.

The project was also transparent about what was included and excluded. The datasets used in the dashboard are sample datasets and do not represent every cooperative, suburb, industry or individual in the full population. This limitation was noted through metadata, report discussion and the presentation.

However, some unintended harm may still arise. Suburbs with higher housing stress could be perceived as “less worth living”, even though the indicator only shows housing pressure in relation to income. Cooperatives excluded by filters could be wrongly seen as completely irrelevant. Community focus labels could also be misunderstood as excluding other groups. Approximate housing cooperative pins may be mistaken for exact locations, and ranking or colour categories may unintentionally stigmatise suburbs, sectors or cooperatives if interpreted without context.

6. CONCLUSION AND FUTURE WORK

This project shows how Mercury's existing Co-op Map can be expanded into a broader visual data exploration environment that supports stakeholder groups with their decision-making needs.

6.1. Recommendation for future use of the visual data exploration environment beyond the project

For cooperative leaders and sector stakeholders, it can support ongoing exploration of cooperative development opportunities, collaboration possibilities and workforce-related issues. For education and employability stakeholders, it can support the search for potential partnerships with cooperatives by allowing users to explore sector relevance, value alignment and accessibility. For housing-related stakeholders and community users, it can support exploration of housing cooperatives, comparison of existing options and identification of potential areas where future housing cooperatives can be established.

6.2. Recommendation for future management of the visual data exploration environment beyond the project

- **Keep datasets current through regular reviews and updates:** Many datasets used in this project, for example, ABS business exits and entries, job vacancies and unemployment rates are updated periodically by their original providers. Ongoing reviews and updates as new releases become available help ensure the visualisation environment reflects current context with relevance and accuracy rather than a historical snapshot.
- **Maintain visibility of metadata and assumptions:** Keeping metadata and assumptions visible helps users correctly interpret proxy measures, such as innovation precinct proximity and housing stress, and use the dashboard more responsibly.

6.3. Possible further improvement and innovation

- **Scale from data subsets to broader directory:** Due to project scope and time constraints, this project used 80 cooperatives as a selected subset. Sample data for NSW innovation precincts and housing cooperatives were also used for Persona 2 and Persona 3. Future versions could expand the cooperative dataset to the full directory and increase coverage across states and territories. This would improve representativeness and allow users to explore a larger section of the Australian cooperative sector.
- **Verify manually coded attributes:** Future versions could improve the reliability of manually harvested fields such as sector, ICA principles, commodity type and cooperative function. If cooperative self-reported data and official classification sources become available, these fields could be verified and presented more consistently.
- **Expand and integrate future datasets:** Further development could incorporate additional datasets that enrich understanding of the cooperative ecosystem. For sector stakeholders, labour shortage and internal membership data could provide deeper insight into cooperative development, expansion or collaboration. For employability stakeholders, internship, skill demand or student survey data could strengthen partnership exploration. For housing-related stakeholders, housing cooperative capacity or membership requirement data could support more practical comparison of existing housing cooperative options.

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